

EXPERIMENT NO 1

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Class- D15A

Roll no- 04

AIM- Study of Wireless Module

THEORY-

1. **Bluetooth**: Short-range wireless technology used for communication between devices like smartphones, headphones, and speakers. Typically works within a range of 10-100 meters, depending on the version.
2. **Wi-Fi**: Wireless networking technology that provides internet access and local area network (LAN) connectivity. It uses radio waves to transmit data and is common in homes, businesses, and public spaces.
3. **Zigbee**: A low-power, short-range wireless communication standard primarily used for IoT (Internet of Things) devices. It's ideal for applications like home automation, smart meters, and sensors.
4. **LoRa (Long Range)**: A low-power, wide-area network (LPWAN) technology designed for long-range communication with minimal power consumption. It's often used for IoT devices, like smart agriculture, tracking, and remote sensors.
5. **5G**: The fifth generation of cellular network technology, offering significantly faster speeds, lower latency, and the ability to connect many more devices than previous generations. It's used for everything from mobile phones to smart cities and autonomous vehicles.
6. **NFC (Near Field Communication)**: A short-range wireless technology used for secure data transfer, typically within a few centimeters. It's

commonly used for contactless payments, ticketing, and sharing data between devices.

7. **Infrared (IR):** A wireless communication technology that uses infrared light to transmit data. It's commonly found in remote controls, short-range data transfer, and some sensor systems.
8. **Satellite Communication:** Uses satellites in Earth's orbit to relay signals for global communication. It's used in GPS, satellite TV, weather forecasting, and long-range communications, especially in remote areas.
9. **Cellular:** Refers to the network of mobile towers and infrastructure that allow devices like smartphones to communicate wirelessly. Cellular networks use technologies like 3G, 4G, and 5G to provide mobile internet and voice communication.

1. Range of Communication

- **Definition:** The maximum distance over which a device can communicate with another device.
- **Importance:** Determines how far apart devices can be while still maintaining a reliable connection. For example, Bluetooth is suited for short-range connections (a few meters), while satellite communication can cover vast global distances.
- **Factors:** Environmental factors like obstacles, interference, and the frequency used can affect range. For example, low-frequency signals can travel farther but might offer lower data rates.

2. Cost

- **Definition:** The financial expense required for deploying and maintaining the technology.
- **Importance:** Cost impacts both the affordability of devices and infrastructure as well as how widely a technology can be adopted.
- **Factors:** Includes hardware costs, network infrastructure, and operational expenses. For instance, Bluetooth and Wi-Fi are relatively inexpensive, while satellite communication and 5G infrastructure can be quite costly.

3. Propagation Delay

- **Definition:** The time it takes for a signal to travel from the sender to the receiver.
- **Importance:** Low propagation delay is crucial for real-time applications like voice and video calls, or autonomous vehicles. High propagation delay can result in lag and poor user experience.
- **Factors:** Affected by the distance between devices (longer distances result in higher delays), and the type of medium (optical fibers, air, space, etc.). For example, satellite communication has high propagation delay due to the long distance to space, while Wi-Fi and Bluetooth have low propagation delays.

4. Power Consumption

- **Definition:** The amount of electrical energy a device consumes during communication.
- **Importance:** Low power consumption is important for battery-powered devices or IoT devices that need to run for extended periods without charging. For example, Zigbee and LoRa are designed to be power-efficient, while 5G might consume more power due to its high throughput capabilities.
- **Factors:** Dependent on the technology's design and usage. For example, Wi-Fi requires more power for high-speed data transmission, while Bluetooth Low Energy (BLE) is optimized for minimal power use.

5. Throughput

- **Definition:** The rate at which data is transmitted over the network, typically measured in bits per second (bps).
- **Importance:** Throughput determines how much data can be sent in a given amount of time, which is key for applications that require large amounts of data to be transferred quickly, such as streaming video or large file transfers.
- **Factors:** Higher frequencies generally offer higher throughput but can have shorter range or greater susceptibility to interference. For example, 5G has extremely high throughput (up to 10 Gbps), while Zigbee offers much lower throughput (250 kbps).

6. Fanout

- **Definition:** The number of devices that can be supported by a single network node or access point.
- **Importance:** Indicates how scalable and capable a technology is in handling multiple connections simultaneously. Technologies with high fanout are suitable for densely populated areas or applications where many devices need to communicate, such as IoT systems.
- **Factors:** This can depend on the technology's network architecture (e.g., mesh vs. star), the capacity of the access points or gateways, and the ability to manage interference. Zigbee, for example, supports a high fanout through its mesh network, whereas Bluetooth and NFC have low fanout due to their short range and limited capacity for simultaneous connections.

Example Comparison with Real-World Use:

- **Range:**
 - *Bluetooth* (short range) is ideal for connecting your wireless headphones to your phone, while *Satellite* (global range) is used for communication in remote areas or for GPS.
- **Cost:**
 - *NFC* is very inexpensive, used for contactless payments, while building and maintaining *5G* infrastructure is significantly more costly due to its need for new towers and equipment.

- **Propagation Delay:**

- 5G has low propagation delay, making it ideal for real-time apps like virtual reality, whereas *satellite communication* experiences noticeable delay due to the long distance the signal must travel.

- **Power Consumption:**

- *LoRa* and *Zigbee* are designed for very low power, ideal for IoT devices that need to run on batteries for years, while *Wi-Fi* and 5G typically consume more power, as they support higher data rates and greater coverage.

- **Throughput:**

- *Wi-Fi* and 5G can handle high data rates, so they're suited for applications like streaming video or downloading large files, whereas *Zigbee* has lower throughput and is used for simple control tasks, like turning a light on/off in a smart home.

- **Fanout:**

- *Zigbee* has high fanout due to its mesh network capabilities, making it great for large-scale IoT deployments, while *Bluetooth* is limited in fanout, making it suitable for personal devices, like a phone and a few connected peripherals.

Here's a comparison of these wireless technologies based on the specified parameters:

1. Bluetooth

- **Range of Communication:** 10-100 meters (varies by version, e.g., Bluetooth 5 has longer range).
- **Cost:** Low. Bluetooth modules are inexpensive to implement.
- **Propagation Delay:** Low, typically in the millisecond range.
- **Power Consumption:** Low. Bluetooth Low Energy (BLE) is designed for minimal power use.
- **Throughput:** Up to 3 Mbps (Bluetooth 4.x), higher in newer versions (up to 2 Mbps in Bluetooth 5).
- **Fanout:** Limited fanout due to its short range, typically supports communication between a few devices.

2. Wi-Fi

- **Range of Communication:** Typically 30-100 meters indoors, up to 300 meters outdoors (depending on environment and Wi-Fi standard).
- **Cost:** Moderate, with routers and adapters being affordable but higher compared to Bluetooth.
- **Propagation Delay:** Low, typically in the millisecond range.
- **Power Consumption:** Moderate to high, depending on the device and usage.
- **Throughput:** Ranges from 11 Mbps (Wi-Fi 4) to 9.6 Gbps (Wi-Fi 6).

- **Fanout:** Moderate. A Wi-Fi router can support many devices simultaneously (often 10-100 devices depending on the router).

3. Zigbee

- **Range of Communication:** 10-100 meters (based on environment and device).
- **Cost:** Very low, particularly for IoT devices.
- **Propagation Delay:** Low, usually in the millisecond range.
- **Power Consumption:** Very low, designed for low-power IoT devices.
- **Throughput:** Around 250 kbps.
- **Fanout:** High, as Zigbee supports mesh networking, allowing many devices (hundreds to thousands) to be interconnected.

4. LoRa

- **Range of Communication:** 2-15 km in rural areas, 2-5 km in urban settings.
- **Cost:** Low to moderate, with modules being affordable.
- **Propagation Delay:** Low, typically in the millisecond range, but due to long-range, it might experience slight delays in some conditions.
- **Power Consumption:** Very low, designed for battery-powered IoT devices.
- **Throughput:** Low, between 0.3 kbps to 27 kbps.
- **Fanout:** Moderate. LoRaWAN networks can support a large number of devices, but the actual fanout depends on the gateway's capacity.

5. 5G

- **Range of Communication:** 100 meters to several kilometers, depending on the environment and 5G deployment (e.g., mmWave has shorter range, sub-6 GHz has longer range).
- **Cost:** High, as it requires new infrastructure and devices.
- **Propagation Delay:** Very low, around 1 ms in ideal conditions (key for real-time applications).
- **Power Consumption:** Moderate to high, especially in high-throughput scenarios.
- **Throughput:** Up to 10 Gbps (theoretical max in ideal conditions).
- **Fanout:** High. 5G can support millions of devices per square kilometer in dense urban environments.

6. NFC

- **Range of Communication:** Extremely short, typically 4 cm or less.
- **Cost:** Very low, especially for simple applications (e.g., contactless payments).
- **Propagation Delay:** Very low, in the nanosecond range.
- **Power Consumption:** Very low, since devices only need power when in close proximity.
- **Throughput:** Very low, around 424 kbps.
- **Fanout:** Very low. NFC is typically one-to-one communication.

7. Infrared (IR)

- **Range of Communication:** Short, typically 1-5 meters.
- **Cost:** Very low, inexpensive components.
- **Propagation Delay:** Low, typically in the microsecond range.
- **Power Consumption:** Low to moderate, depending on usage.
- **Throughput:** Low, around 1-4 Mbps.
- **Fanout:** Very low. IR is often used for point-to-point communication, like remote controls.

8. Satellite Communication

- **Range of Communication:** Global, as it covers vast distances using satellites in orbit.
- **Cost:** High, both for infrastructure (launching satellites) and user equipment.
- **Propagation Delay:** High, typically 100-500 ms due to the distance the signal must travel.
- **Power Consumption:** Moderate to high, depending on the satellite and ground station.
- **Throughput:** Moderate to high, but varies greatly with satellite type; can be up to several Gbps in modern systems.
- **Fanout:** Very high. A single satellite can serve millions of users globally.

9. Cellular (2G/3G/4G/5G)

- **Range of Communication:** Varies by generation (e.g., 4G has a range of a few kilometers, 5G's mmWave range is much shorter).
- **Cost:** Moderate to high, as infrastructure and devices can be costly.
- **Propagation Delay:** Low to moderate, with 5G offering ultra-low latency (~1 ms) and older generations having higher delays.
- **Power Consumption:** Varies, with newer generations (like 5G) being more power-efficient but requiring more energy for higher data rates.
- **Throughput:** Ranges from 100 kbps (2G) to 10 Gbps (5G).
- **Fanout:** Very high, especially in urban areas. Cellular towers can support thousands of users simultaneously.

Summary Table:

Technology	Range	Cost	Propagation Delay	Power Consumption	Throughput	Fanout
Bluetooth	10-100 meters	Low	Low	Low	Up to 3 Mbps	Low
Wi-Fi	30-100 meters	Moderate	Low	Moderate	Up to 9.6 Gbps	Moderate
Zigbee	10-100 meters	Very Low	Low	Very Low	250 kbps	High
LoRa	2-15 km	Low	Low	Very Low	0.3-27 kbps	Moderate
5G	100 meters - 3 km	High	Very Low	Moderate-High	Up to 10 Gbps	High
NFC	4 cm	Very Low	Very Low	Very Low	424 kbps	Very Low
Infrared (IR)	1-5 meters	Very Low	Low	Low	1-4 Mbps	Very Low

Satellite	Global	High	High	Moderate-High	Moderate-High	Very High
Cellular	Few km-100 km	Moderate-High	Low-Moderate	Moderate	100 kbps-10 Gbps	Very High

Each technology excels in different areas, with trade-offs between range, cost, power consumption, throughput, and scalability depending on the application.